The documentation and process conversion measures necessary to comply with this revision shall be completed by 13 April 1999.

INCH-POUND

MIL-PRF-19500/477C
13 January 1999
SUPERSEDING
MIL-S-19500/477B
12 June 1982

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, FAST RECOVERY, POWER RECTIFIER,
TYPES 1N5802, 1N5804, 1N5806, 1N5807, 1N5809, AND 1N5811,
1N5802US, 1N5804US, 1N5806US, 1N5807US, 1N5809US, AND 1N5811US
JAN, JANTX, JANTXV, JANJ, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, fast recovery, power rectifier diodes. Five levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two level of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figures 1 through 8.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

1.3.1 Ratings applicable to all Part or Identifying Numbers (PIN).

T_{STG} = -65°C to +175°C.
T_{op} = -65°C to +175°C.
$T_{J(max)}$ = +175°C.

1.3.2 Ratings applicable to individual types.

Types	V_{RWM}	I_{O1} T_L = +75°C L = .375" (9.52 mm) 1/	I_{O2} T_A = +55°C 2/	I_{FSM} at +25°C Operating at I_{O2} t_p = 8.3 ms	t_{rr}	$R_{\theta JL}$ at L = .375" (9.52 mm)	$R_{\theta JEC}$ 3/
1N5802, 1N5802US	50	2.5 A 4/	1.0 A 5/	35 A(pk)	25 ns	36°C/W	20°C/W
1N5804, 1N5804US	100	2.5 A 4/	1.0 A 5/	35 A(pk)	25 ns	36°C/W	20°C/W
1N5806, 1N5806US	150	2.5 A 4/	1.0 A 5/	35 A(pk)	25 ns	36°C/W	20°C/W
1N5807, 1N5807US	50	6.0 A 6/	3.0 A 7/	125 A(pk)	30 ns	22°C/W	10°C/W
1N5809, 1N5809US	100	6.0 A 6/	3.0 A 7/	125 A(pk)	30 ns	22°C/W	10°C/W
1N5811, 1N5811US	150	6.0 A 6/	3.0 A 7/	125 A(pk)	30 ns	22°C/W	10°C/W

1/ $T_{EC} = T_L$ at L = 0 or $T_{end\ cap}$ for US suffix devices.
2/ This rating is typical for PC boards where thermal resistance from mounting point to ambient is sufficiently controlled where T_{OP} and $T_{J(max)}$ in paragraph 1.3 are not exceeded.
3/ US suffix devices only.
4/ Derate at 25 mA/°C for T_L above +75°C.
5/ Derate at 8.33 mA/°C for T_A above +55°C.
6/ Derate at 60 mA/°C for T_L above +75°C.
7/ Derate at 25 mA/°C for T_A above +55°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25°C.

Types	V_{BR}	I_{R1} at V_R = V_{RWM} T_A = +25°C	I_{R2} at V_R = V_{RWM} T_A = +100°C
	(V dc)	µA dc	µA dc
1N5802, 1N5802US	60	1.0	50
1N5804, 1N5804US	110	1.0	50
1N5806, 1N5806US	160	1.0	50
1N5807, 1N5807US	60	5.0	150
1N5809, 1N5809US	110	5.0	150
1N5811, 1N5811US	160	5.0	150

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

V_{fr} . . . . . Forward recovery voltage.
$I_{(BR)}$ . . . . . Current for testing breakdown voltage.

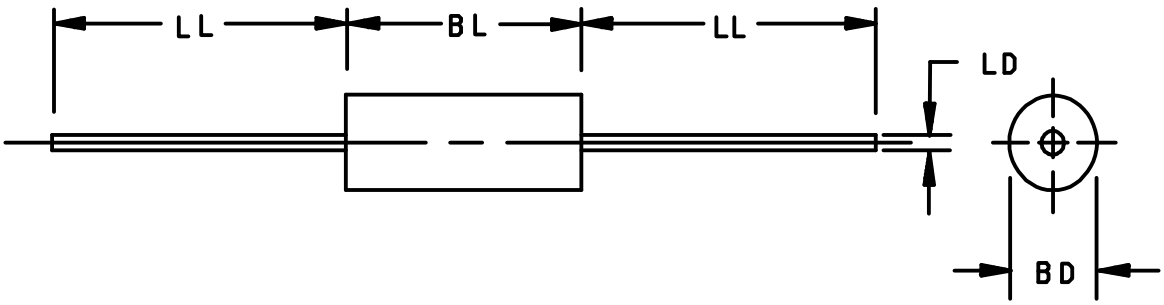
3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1 through 8 herein.

3.3.1 Diode construction. These devices shall be constructed utilizing non-cavity double plug construction with high temperature metallurgical bonding between both sides of the silicon die and terminal pins (see MIL-PRF-19500). Metallurgical bond shall be in accordance with the requirements of category I in MIL-PRF-19500. US version devices shall be structurally identical to the non-surface mount devices except for lead terminations.

3.3.2 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein.

3.4 Marking. Devices shall be marked as specified in MIL-PRF-19500.

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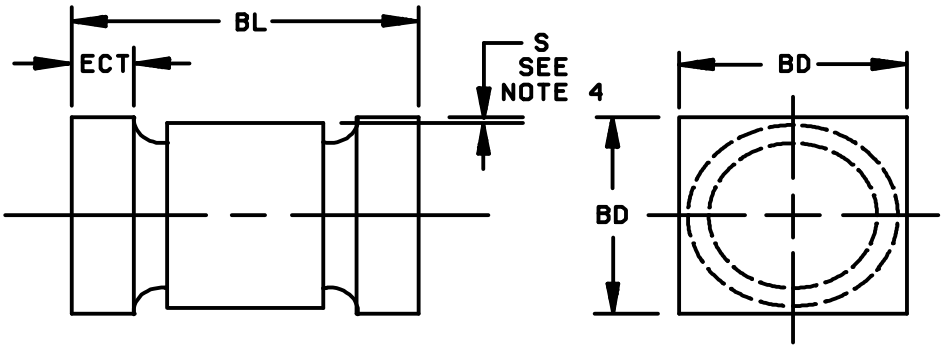


Ltr.	Dimensions								Notes
	1N5802, 1N5804, 1N5806				1N5807, 1N5809, 1N5811				
	Inches		Millimeters		Inches		Millimeters		
	Min	Max	Min	Max	Min	Max	Min	Max	
BD	.065	.085	1.65	2.16	.115	.165	2.92	4.19	4
BL	.125	.250	3.18	6.35	.130	.300	3.30	7.62	3
LD	.027	.032	0.69	0.81	.037	.042	0.94	1.07	3
LL	.700	1.30	17.78	33.02	.900	1.30	22.86	33.02	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension LD shall include the sections of the lead over which the diameter is uncontrolled. This uncontrolled area is defined as the zone between the edge of the diode body and extending .050 inch (1.27 mm) onto the leads.
4. Dimension BD shall be measured at the largest diameter.
5. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions.

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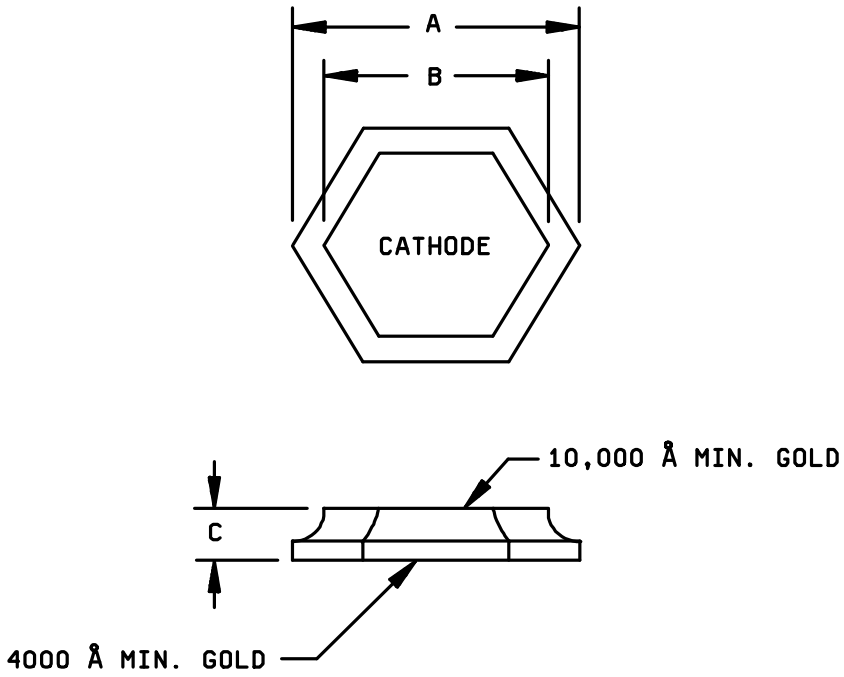


Ltr.	Dimensions								Notes
	D-5A 1N5802US, 1N5804US, 1N5806US				D-5B 1N5807US, 1N5809US, 1N5811US				
	Inches		Millimeters		Inches		Millimeters		
	Min	Max	Min	Max	Min	Max	Min	Max	
BD	.091	.103	2.31	2.62	.137	.148	3.48	3.76	
BL	.168	.200	4.27	5.08	.200	.225	5.08	5.72	
ECT	.019	.028	0.48	0.71	.019	.028	0.48	0.71	
S	.003		0.80		.003		0.80		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Gap not controlled, shape of body and gap not controlled.
4. Dimensions are pre-solder dip.
5. Cathode marking to be either in color band, three dots spaced equally, or a color dot on the face of the end cap.
6. Color dots will be .020 inch (0.51 mm) diameter minimum and those on the face of the end cap shall not lie within .020 inch (0.51 mm) of the mounting surface.
7. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions of surface mount family.

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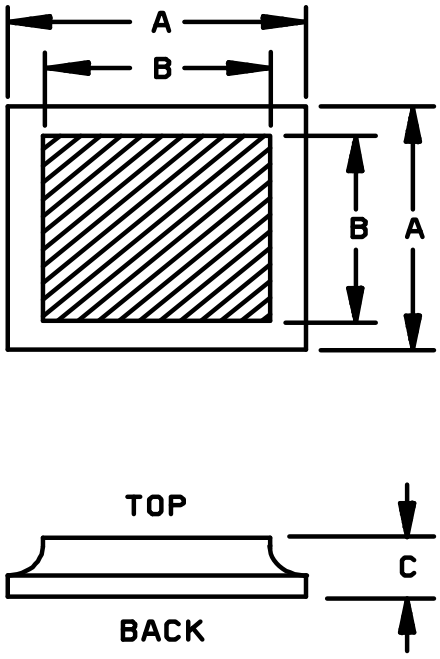


Ltr.	Dimensions								Notes
	1N5802, 1N5804, 1N5806				1N5807, 1N5809, 1N5811				
	Inches		Millimeters		Inches		Millimeters		
	Min	Max	Min	Max	Min	Max	Min	Max	
A	.047	.053	1.19	1.35	.085	.091	2.16	2.31	
B	.033	.037	0.84	0.94	.072	.076	1.83	1.93	
C	.007	.011	0.18	0.28	.007	.011	0.18	0.28	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. JANC (A-version) die dimensions.

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Ltr.	Dimensions								Notes
	1N5802, 1N5804, 1N5806				1N5807, 1N5809, 1N5811				
	Inches		Millimeters		Inches		Millimeters		
	Min	Max	Min	Max	Min	Max	Min	Max	
A	.031	.037	0.79	0.94	.062	.068	1.57	1.73	
B	.017	.023	0.43	0.58	.050	.056	1.27	1.42	
C	.008	.012	0.20	0.30	.008	.012	0.20	0.30	

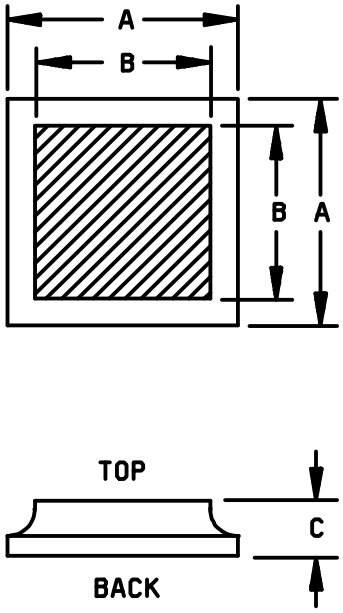
Metallization:

Top: (Anode). . . . . Aluminum
Back (Cathode). . . . . Gold

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. JANC (B-version) die dimensions.

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Ltr.	Dimensions								Notes
	1N5802, 1N5804, 1N5806				1N5807, 1N5809, 1N5811				
	Inches		Millimeters		Inches		Millimeters		
	Min	Max	Min	Max	Min	Max	Min	Max	
A	.031	.037	0.79	0.94	.062	.068	1.57	1.73	
B	.017	.023	0.43	0.58	.050	.056	1.27	1.42	
C	.008	.012	0.20	0.30	.008	.012	0.20	0.30	

DESIGN DATA

Metallization:

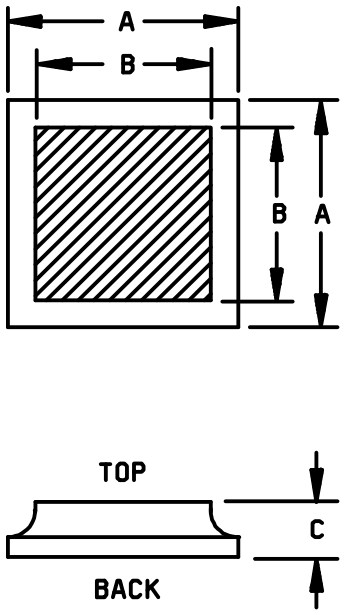
Top: (Anode). . . . . Aluminum
Back (Cathode). . . . . Silver

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 5. JANC (C-version) die dimensions.

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Ltr.	Dimensions								Notes
	1N5802, 1N5804, 1N5806				1N5807, 1N5809, 1N5811				
	Inches		Millimeters		Inches		Millimeters		
	Min	Max	Min	Max	Min	Max	Min	Max	
A	.031	.037	0.79	0.94	.062	.068	1.57	1.73	
B	.017	.023	0.43	0.58	.050	.056	1.27	1.42	
C	.008	.012	0.20	0.30	.008	.012	0.20	0.30	

DESIGN DATA

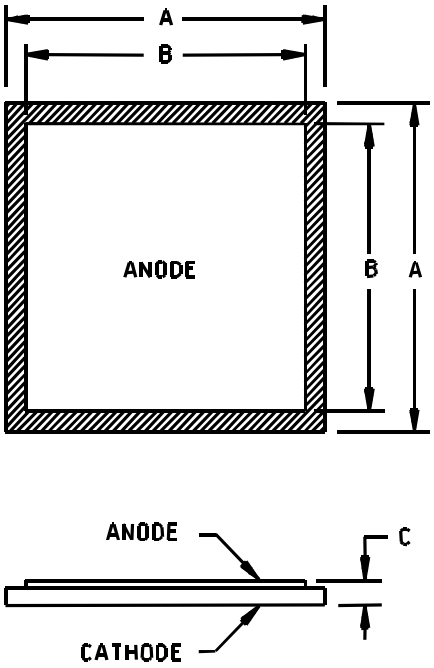
Metallization:

Top: (Anode). . . . . Silver
Back (Cathode). . . . . Silver

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 6. JANC (D-version) die dimensions.

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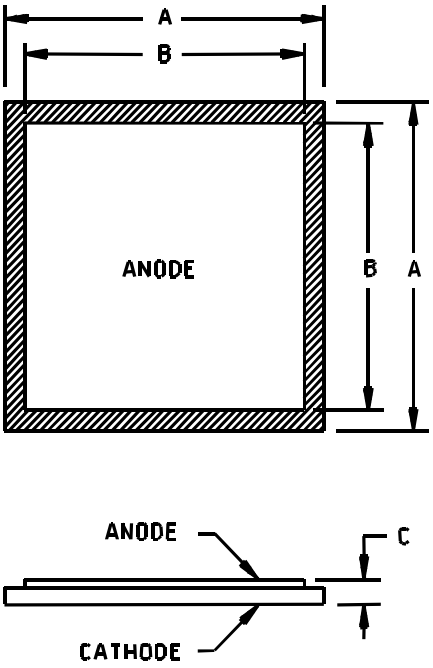
1N5802, 1N5804, 1N5806

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	1.00	1.09
B	.031	.035	0.79	0.89
C	.008	.012	0.20	0.30

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Anode is aluminum at 45,000 Å minimum.
4. Cathode is gold at 2500 Å minimum.

FIGURE 7. JANC (E-version) die dimensions.

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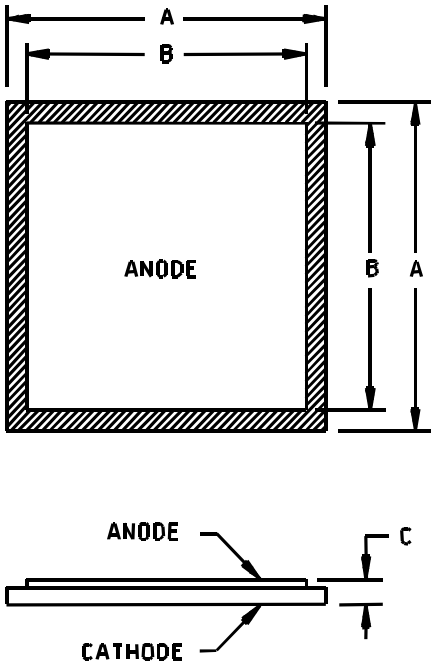
1N5807, 1N5809, 1N5811

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.066	.070	1.68	1.78
B	.057	.061	1.45	1.55
C	.008	.012	0.20	0.30

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Anode is aluminum at 60,000 Å minimum.
4. Cathode is gold at 2500 Å minimum.

FIGURE 7. JANC (E-version) die dimensions - Continued.

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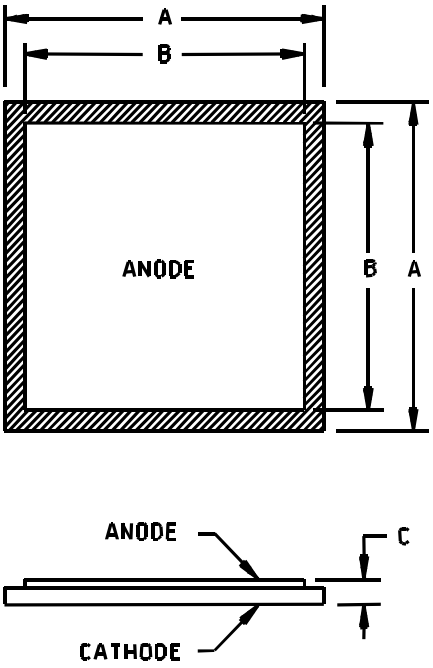
1N5802, 1N5804, 1N5806

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	1.00	1.09
B	.031	.035	0.79	0.89
C	.008	.012	0.20	0.30

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Anode is aluminum at 45,000 Å minimum.
4. Cathode is silver at 2500 Å minimum.

FIGURE 8. JANC (F-version) die dimensions.

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1N5807, 1N5809, 1N5811

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.066	.070	1.68	1.78
B	.057	.061	1.45	1.55
C	.008	.012	0.20	0.30

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Anode is aluminum at 60,000 Å minimum.
4. Cathode is silver at 2500 Å minimum.

FIGURE 8. JANC (F-version) die dimensions - Continued.

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3.4.1 Marking for US devices. For US version devices only, all marking may be omitted from the device except for the cathode marking. All marking which is omitted from the body of the device shall appear on the label of the initial container.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. For JANJ level, see 3.3.1 through 3.3.1.3 of MIL-PRF-19500. Supplier imposed requirements shall be documented in the QM plan and must be submitted to the Qualifying Activity for approval. Radiation characterization may be submitted in the QM plan at the option of the manufacturer.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal impedance (see 4.5.3)	Thermal impedance (see 4.5.3)
9	I_{R1} and V_{FM1}	Not applicable
11 2/	I_{R1} and V_{FM1}; ΔI_R ± 100 percent of initial reading or ±150 nA dc(1N5802, 1N5804, 1N5806) or ±500 nA dc (1N5807, 1N5809, 1N5811), whichever is greater. $\Delta V_{FM1} \leq \pm 0.05$ V dc.	I_{R1} and V_{FM1}
12	See 4.3.1	See 4.3.1
13 2/, 3/	Subgroups 2 and 3 of table I herein; $\Delta I_R \leq 100$ percent of initial reading or ± 150 nA dc (1N5802, 1N5804, 1N5806) or ± 500 nA dc (1N5807, 1N5809, 1N5811), whichever is greater. $\Delta V_{FM1} \leq \pm 0.05$ V dc. Scope-display evaluation (see 4.5.4).	Subgroup 2 of table I herein;ΔI_{R1} ± 100 percent of initial reading or ± 250 nA dc (1N5802, 1N5804, 1N5806) or ± 1 µA dc (1N5807, 1N5809, 1N5811), whichever is greater. $\Delta V_{FM1} \leq \pm 0.05$ V dc. Scope-display evaluation (see 4.5.4).

1/ Thermal impedance shall be performed any time after screen 3.
2/ Also applies to “US” suffix devices.
3/ Except thermal impedance, if already performed.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:
With the written approval of the qualifying activity and the preparing activity, an alternate burn-in flow may be used (I_O or I_F). A study illustrating equivalence with ACOL testing will be required.

T_A = Room ambient as defined in 4.5 of MIL-STD-750.
I_O = 1.0 A (1N5802, 1N5802US, 1N5804, 1N5804US, 1N5806, 1N5806US)
I_O = 3.0 A (1N5807, 1N5807US, 1N5809, 1N5809US, 1N5811, 1N5811US)
V_R = rated V_{RWM} (see 1.3.2), f = 50-60 Hz.

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4.3.1.1 Alternate mounting conditions (For -US devices only). At the option of the manufacturer, any clips or heat sink mounting configurations may be utilized provided that one of the following conditions be met:

a. T_{EC} = +75°C to +125°C, $V_R = V_{RWM}$ rated, f = 50-60 Hz, I_O = rated (see 1.3.2).

b. Temporary attachment of leads or equivalent (thermal properties not to exceed the leaded part) T_J shall not exceed 200°C , f = 50-60 Hz. T_J = +150°C ±25°C, $I_O = I_{O2(rated)}$, $V_R = V_{RWM}$ rated.

4.3.2 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-PRF-19500.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein except $Z_{\theta JX}$ need not to be performed. See 4.5.2 for delta limits (JANS only).

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
3	4066	I_{FSM} = rated (see 1.3); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on $I_O = I_{02}$ rated (see 1.3); V_{RSM} = rated (see 1.3); T_A = room ambient as defined in 4.5 of MIL-STD-750.
4	1037	$I_O = I_{O2}$ rated (see 1.3.2 and 4.3.1); V_R = rated V_{RWM} (see 1.3.2 and 4.3.1); f = 60 Hz for a minimum of 2,000 cycles. T_L => +55°C at .375 inch.
5	1027	$I_O \geq I_{O2}$ rated minimum (see 1.3.2); V_R = 0 ; f = 60 Hz, T_A = +150°C minimum. Adjust T_A or I_O to achieve an average $T_J \geq$ +250°C.
6	3101 or 4081	See 4.5.1 and figure 8; maximum $R_{\theta JL}$ shall be as specified on table IV herein.

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV of MIL-PRF-19500).

Subgroup	Method	Condition
2	4066	I_{FSM} = rated (see 1.3); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on $I_O = I_{O2}$ rated (see 1.3); V_{RSM} = rated (see 1.3). T_A = +25°C.
3	1027	f = 60 Hz; $V_R = V_{RWM}$ rated (see 1.3.2 and 4.3.1) T_A = room ambient as defined in 4.5 of MIL-STD-750. Adjust I_O to achieve T_J = 150°C min. T_J max shall not exceed 200°C.
5		Not applicable.
6	1032	T_A = +175°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein except $Z_{\theta JX}$ need not to be performed. See 4.5.2 for delta limits (JANS only).

Subgroup	Method	Condition
2	2036	NOTE: Not applicable for US types. Tension: condition A, 4 pounds - 1N5802, 1N5804, 1N5806 5 pounds - 1N5807, 1N5809, 1N5811 Fatigue: Condition E, 2 pounds.
3		Not applicable.
6	1026	$I_O = I_{O2}$ rated; f = 60 Hz; $V_R = V_{RWM}$ rated (see 1.3.2 and 4.3.1).

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Thermal resistance. Thermal resistance shall be measured in accordance with MIL-STD-750, method 3101 or 4081. Read and record data in accordance with group E herein shall be included in the qualification report. Forced moving air or draft shall not be permitted across the device during test. The maximum limit for $R_{\theta JL}$ or $R_{\theta JEC}$ under these test conditions shall be as specified in 1.3.2. The following conditions shall apply:

1N5802, 1N5804, 1N5806	1N5807, 1N5809, 1N5811
I_H = 2.0 A dc minimum	2.0 A dc minimum
I_M = 10 mA	10 mA
t_{MD} = 250 µs maximum	250 µs maximum

The device shall be allowed to reach thermal equilibrium at current I_H before the measurement shall be made.
Lead spacing: L = .375 inch (9.52 mm) for leaded devices
L = 0 (endcap mount) for -US devices.

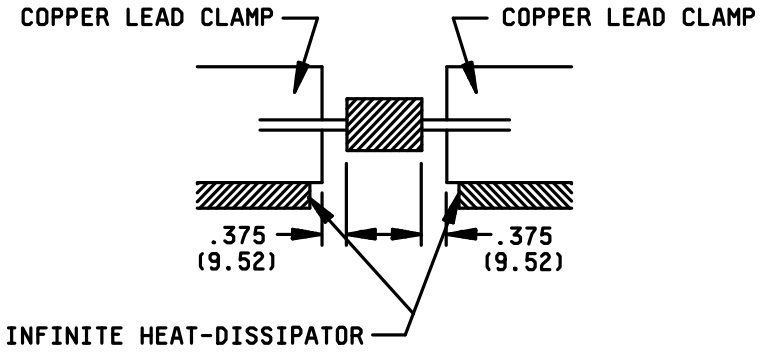


FIGURE 9. Mounting arrangement.

4.5.2 Delta Requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1	Forward voltage 1N5802,1N5804,1N5806 1N5807,1N5809,1N5811	4011	Duty cycle ≤ 2 percent (pulsed); t_p = 8.3 ms (max) I_{FM} = 1.0 A(pk) I_{FM} = 4.0 A(pk)	ΔV_{F1}	± 50 mV dc change from of initial value	
2	Reverse current 1N5802,1N5802US 1N5804,1N5804US 1N5806,1N5806US 1N5807,1N5807US 1N5809,1N5809US 1N5811,1N5811US	4016	DC method V_R = 50 V dc V_R = 100 V dc V_R = 150 V dc V_R = 50 V dc V_R = 100 V dc V_R = 150 V dc	ΔI_{R1}	100% or ±150 nA dc change from initial reading, whichever is greater. 100% or ±500 nA dc change from initial reading, whichever is greater.	

1/ Devices which exceed the group A limits for this test shall not be accepted.

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TABLE I. Group A inspection .

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical Examination	2071					
Subgroup 2						
Thermal impedance 2/	3101	see 4.5.3	$Z_{\theta JX}$			
1N5802, 1N5802US 1N5804, 1N5804US 1N5806, 1N5806US					4.5	°C/W
1N5807, 1N5807US 1N5809, 1N5809US 1N5811, 1N5811US					1.5	°C/W
Forward voltage	4011	Duty cycle ≤ 2 percent (pulsed); t_p = 8.3 ms (max)	V_{FM1}		0.875	V (pk)
1N5802, 1N5802US 1N5804, 1N5804US 1N5806, 1N5806US		I_{FM} = 1.0 A(pk)				
1N5807, 1N5807US 1N5809, 1N5809US 1N5811, 1N5811US		I_{FM} = 4.0 A(pk)				
Forward voltage	4011	Duty cycle ≤ 2 percent (pulsed); t_p = 8.3 ms (max)	V_{FM2}		0.975	V (pk)
1N5802, 1N5802US 1N5804, 1N5804US 1N5806, 1N5806US		I_{FM} = 2.5 A(pk)				
1N5807, 1N5807US 1N5809, 1N5809US 1N5811, 1N5811US		I_{FM} = 6.0 A(pk)				
Reverse current	4016	DC method	I_{R1}			
1N5802, 1N5802US		V_R = 50 V dc			1.0	µA dc
1N5804, 1N5804US		V_R = 100 V dc			1.0	µA dc
1N5806, 1N5806US		V_R = 150 V dc			1.0	µA dc
1N5807, 1N5807US		V_R = 50 V dc			5.0	µA dc
1N5809, 1N5809US		V_R = 100 V dc			5.0	µA dc
1N5811, 1N5811US		V_R = 150 V dc			5.0	µA dc
Breakdown voltage	4021	$I_{(BR)}$ = 100 µA dc	$V_{(BR)1}$			
1N5802, 1N5802US 1N5807, 1N5807US				60		V dc
1N5804, 1N5804US 1N5809, 1N5809US				110		V dc
1N5806, 1N5806US 1N5811, 1N5811US				160		V dc

See footnotes at end of table.

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TABLE I. Group A inspection .

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current		DC method	I_{R2}			
1N5802, 1N5802US		V_R = 50 V dc			50	µA dc
1N5804, 1N5804US		V_R = 100 V dc			50	µA dc
1N5806, 1N5806US		V_R = 150 V dc			50	µA dc
1N5807, 1N5807US		V_R = 50 V dc			150	µA dc
1N5809, 1N5809US		V_R = 100 V dc			150	µA dc
1N5811, 1N5811US		V_R = 150 V dc			150	µA dc
Forward voltage	4011	Duty cycle ≤ 2 percent (pulsed); t_p = 8.3 ms (max)	V_{FM3}		0.800	V (pk)
1N5802, 1N5802US 1N5804, 1N5804US 1N5806, 1N5806US		I_{FM} = 1.0 A(pk)				
1N5807, 1N5807US 1N5809, 1N5809US 1N5811, 1N5811US		I_{FM} = 4.0 A(pk)				
Low-temperature operation:		T_A = -65°C				
Forward voltage	4011	Duty cycle ≤ 2 percent (pulsed); t_p = 8.3 ms (max)	V_{FM4}		1.075	V (pk)
1N5802, 1N5802US 1N5804, 1N5804US 1N5806, 1N5806US		I_{FM} = 1.0 A(pk)				
1N5807, 1N5807US 1N5809, 1N5809US 1N5811, 1N5811US		I_{FM} = 4.0 A(pk)				
Breakdown voltage	4021	$I_{(BR)}$ = 100 µA dc	$V_{(BR)2}$			
1N5802, 1N5802US 1N5807, 1N5807US				60		V dc
1N5804, 1N5804US 1N5809, 1N5809US				100		V dc
1N5806, 1N5806US 1N5811, 1N5811US				150		V dc
Subgroup 4						
Reverse recovery time	4026	Condition B	t_{rr}			
1N5802, 1N5802US 1N5804, 1N5804US 1N5806, 1N5806US		$I_F = I_R$ = 0.5 A I_{RM} (REC) = 0.5 A(pk) di/dt = 65 A/µs (min)			25	ns
1N5807, 1N5807US 1N5809, 1N5809US 1N5811, 1N5811US		$I_F = I_R$ = 1.0 A I_{RM} (REC) = 0.1 A(pk) di/dt = 100 A/µs (min)			30	ns

See footnotes at end of table.

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TABLE I. Group A inspection .

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 – continued						
Capacitance	4001	V_R = 10 V; f = 1 Mhz; V_{sig} = 50 mV (p-p)	C_J			
1N5802, 1N5802US 1N5804, 1N5804US 1N5806, 1N5806US					25	pF
1N5807, 1N5807US 1N5809, 1N5809US 1N5811, 1N5811US					60	pF
Forward recovery voltage 1N5802, 1N5802US 1N5804, 1N5804US 1N5806, 1N5806US	4026	t_r = 8 ns I_{FM} = 250 mA	V_{FRM}		2.2	V (pk)
1N5807, 1N5807US 1N5809, 1N5809US 1N5811, 1N5811US		I_{FM} = 500 mA				
Forward recovery time 1N5802, 1N5802US 1N5804, 1N5804US 1N5806, 1N5806US	4026	$t_p \geq 20$ ns, t_r = 8 ns, the test is measured at V_{FR} = 1.1 x V_F I_{FM} = 250 mA	t_{fr}		15	ns
1N5807, 1N5807US 1N5809, 1N5809US 1N5811, 1N5811US		I_{FM} = 500 mA				
Subgroup 5, 6, 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ Not applicable to JANHC and JANKC devices.

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			32 devices c = 0
Thermal shock (glass strain)	1056	500 cycles; condition A	
Hermetic seal Electrical measurement	1071	See table I, group A, subgroup 2, except $Z_{\theta JX}$ need not to be performed.	
Subgroup 2			32 devices c = 0
Blocking life	1048	t = 1,000 hours; T_A = +150°C; V_R DC = 80 - 85 percent rated V_{RWM} (see 1.3.2)	
Electrical measurement		See table I, group A, subgroup 2, except $Z_{\theta JX}$ need not to be performed.	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance, junction to lead	3101 or 4081	See 4.5.1	

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4.5.3 Thermal impedance. Thermal impedance $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit for $Z_{\theta JX}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of Statistical Process Control and applied in screening of all subsequent lots. This limit shall not exceed the group A, subgroup 2 limit. The following conditions shall apply:

I_H = 5 A minimum.
t_H = 10 ms.
I_M = 1 mA to 10 mA.
t_{MD} = 100 µs maximum.

4.5.3.1 For initial qualification and requalification. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum) prior to shipment. Twenty-two samples shall be serialized and provided to the qualifying activity for test correlation.

4.5.4 Scope-display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 50 to 100 µA per division and 20 to 50 V per division. Reverse current over the knee shall be at 500 µA minimum. Any discontinuity or dynamic instability of the trace shall be cause for rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.2).

c. Product assurance level, type designator, and for die acquisition, the JANHC and JANKC identification (see figures 3 through 8)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

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6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (example; JANHCA1N5802) will be identified on the QPL.

JANC ordering information						
PIN	Manufacturer					
	14552	12969	12969	12969	33178	
1N5802	JANHCA1N5802 JANKCA1N5802	JANHCB1N5802 JANKCB1N5802	JANHCC1N5802 JANKCC1N5802	JANHCD1N5802 JANKCD1N5802	JANHCE1N5802 JANKCE1N5802	JANHCF1N5802 JANKCF1N5802
1N5804	JANHCA1N5804 JANKCA1N5804	JANHCB1N5804 JANKCB1N5804	JANHCC1N5804 JANKCC1N5804	JANHCD1N5804 JANKCD1N5804	JANHCE1N5804 JANKCE1N5804	JANHCF1N5804 JANKCF1N5804
1N5806	JANHCA1N5806 JANKCA1N5806	JANHCB1N5806 JANKCB1N5806	JANHCC1N5806 JANKCC1N5806	JANHCD1N5806 JANKCD1N5806	JANHCE1N5806 JANKCE1N5806	JANHCF1N5906 JANKCF1N5806
1N5807	JANHCA1N5807 JANKCA1N5807	JANHCB1N5807 JANKCB1N5807	JANHCC1N5807 JANKCC1N5807	JANHCD1N5807 JANKCD1N5807	JANHCE1N5807 JANKCE1N5807	JANHCF1N5807 JANKCF1N5807
1N5809	JANHCA1N5809 JANKCA1N5809	JANHCB1N5809 JANKCB1N5809	JANHCC1N5809 JANKCC1N5809	JANHCD1N5809 JANKCD1N5809	JANHCE1N5809 JANKCE1N5809	JANHCF1N5809 JANKCF1N5809
1N5811	JANHCA1N5811 JANKCA1N5811	JANHCB1N5811 JANKCB1N5811	JANHCC1N5811 JANKCC1N5811	JANHCD1N5811 JANKCD1N5811	JANHCE1N5811 JANKCE1N5811	JANHCF1N5811 JANKCF1N5811

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-1913)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/477C	2. DOCUMENT DATE 99/01/13

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, FAST RECOVERY, POWER RECTIFIER, TYPES 1N5802, 1N5804, 1N5806, 1N5807, 1N5809, AND 1N5811, 1N5802US, 1N5804US, 1N5806US, 1N5807US, 1N5809US, AND 1N5811US JAN, JANTX, JANTXV, JANJ, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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